

Python Programming Fundamentals

Lists of Dictionaries

Programming Fundamentals Team

SETU

2026-10-13



Outline

1. A List of Records
2. Displaying Records
3. Searching
4. Filtering
5. Adding New Records



Outline

1. A List of Records
2. Displaying Records
3. Searching
4. Filtering
5. Adding New Records



1. A List of Records

```
movies = [
    {"title": "Jaws", "year": 1975, "rating": 5},
    {"title": "Alien", "year": 1979, "rating": 5},
    {"title": "Shrek", "year": 2001, "rating": 4}
]
```

1. A List of Records

```
movies = [
    {"title": "Jaws", "year": 1975, "rating": 5},
    {"title": "Alien", "year": 1979, "rating": 5},
    {"title": "Shrek", "year": 2001, "rating": 4}
]
```

Each dictionary is one record.

The list stores the collection.



Outline

1. A List of Records
2. Displaying Records
3. Searching
4. Filtering
5. Adding New Records



2. Displaying Records

```
for movie in movies:
    print(movie["title"], movie["year"])
      #Jaws 1975
      #Alien 1979
      #Shrek 2001
```



2. Displaying Records

```
for movie in movies:
    print(movie["title"], movie["year"])
      #Jaws 1975
      #Alien 1979
      #Shrek 2001
```

Formatted output:

```
for movie in movies:
    print(f"{movie['title']} ({movie['year']})")
    # Shrek 2001
    # Jaws (1975)
    # Alien (1979)
    # Shrek (2001)
```



Outline

1. A List of Records

2. Displaying Records

3. Searching

4. Filtering

5. Adding New Records



3. Searching

```
search_title = input("Enter title: ")

for movie in movies:
    if movie["title"].lower() == search_title.lower():
        print("Found:", movie)

        # e.g. input: Shrek
        # output: Found: {'title': 'Shrek', 'year': 2001, 'rating': 4}
```



3. Searching

```
search_title = input("Enter title: ")

for movie in movies:
    if movie["title"].lower() == search_title.lower():
        print("Found:", movie)

        # e.g. input: Shrek
        # output: Found: {'title': 'Shrek', 'year': 2001, 'rating': 4}
```

Important ideas:

- loop through each record
- check one field (make both search_title and movie[“title”] lowercase for case-insensitive comparison)
- display the matching record



Outline

1. A List of Records

2. Displaying Records

3. Searching

4. Filtering

5. Adding New Records



4. Filtering

```
for movie in movies:
    if movie["rating"] >= 5:
        print(movie["title"])
                # output: Jaws
                # output: Alien
```



4. Filtering

```
for movie in movies:
    if movie["rating"] >= 5:
        print(movie["title"])
                # output: Jaws
                # output: Alien
```

Filtering means selecting only the records that match a condition.



Outline

1. A List of Records

2. Displaying Records

3. Searching

4. Filtering

5. Adding New Records



5. Adding New Records

```
new_movie = {
    "title": input("Title: "),
    "year": int(input("Year: ")),
    "rating": int(input("Rating: "))
}

movies.append(new_movie) # add the new record
```



5. Adding New Records

```
# continued from above
# test by printing the list of movies again
for movie in movies:
    print(movie["title"], movie["year"])
    #assume I entered "Odessey" for title, 2026 for year, and 5 for rating
    # output: Shrek 2001
    # output: Jaws 1975
    # output: Alien 1979
    # output: Odessey 2026
```



5. Adding New Records

```
# continued from above
# test by printing the list of movies again
for movie in movies:
    print(movie["title"], movie["year"])
    #assume I entered "Odessey" for title, 2026 for year, and 5 for rating
    # output: Shrek 2001
    # output: Jaws 1975
    # output: Alien 1979
    # output: Odessey 2026
```

This is an example of the Create step in CRUD.



Thanks for Watching - Any questions?

